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Enhanced degradation of tetracycline (TC) by FeCoNiCuAl high-entropy alloy powders through ball milling

Hong Li,^a Lijuan Zhou,^a Hongjie Nie,^a Junwei Gui,^c Bowen Zhao^{ID}*^b and Zhengwang Zhu^a

High-entropy alloys (HEAs) have garnered significant research interest owing to their exceptional catalytic properties in the degradation of organic pollutants. In this study, we report the synthesis of homogeneous FeCoNiCuAl HEA powders through an integrated single-roller melt-spinning and ball-milling approach. Compared to conventional gas-atomized counterparts, the ball-milled HEA powders demonstrate significantly enhanced catalytic performance in a Fenton-like system toward tetracycline (TC) degradation across varied temperature ranges. Notably, the BM-280 catalyst, processed at 280 rpm, achieves an outstanding TC removal efficiency of 94.8% within 60 min under H₂O₂ activation. The enhanced activity is attributed to the *in situ* generation of nanoscale active sites and a nanoporous structure, which synergistically promote reaction kinetics and interfacial mass transfer. These findings offer a novel and practicable strategy for designing advanced HEA-based advanced catalytic materials toward sustainable environmental purification.

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1. Introduction

The escalating global consumption of antibiotics has posed unprecedented and intractable challenges to aquatic ecosystems.^{1–4} Tetracycline (TC), as a representative refractory organic pollutant, persists and accumulates in water bodies, resulting in severe threats to environmental security and human health.^{5–8} This persistence can foster the proliferation of antibiotic-resistant genes and bacteria, thereby elevating the risk of a public health crisis. Among various remediation strategies, advanced oxidation processes (AOPs) have emerged as some of the most effective approaches, owing to their high efficiency and operational feasibility.^{9–11} Consequently, developing highly efficient and stable catalytic materials for the advanced degradation of such persistent pharmaceutical contaminants in AOP systems represents an urgent and critical research imperative.

High entropy alloys (HEAs) have achieved controllable design of the electronic structure and atomic configuration on the catalyst surface through an almost infinite space of elemental combinations, effectively resolving the long-standing bottlenecks in catalysis related to activity, selectivity, and stability.^{12–16} Thus, the HEA materials based on their inherently disordered chemical structures have garnered intensive

attention as an innovative catalyst in various applications, including the hydrogen evolution reaction (HER), oxygen evolution reaction (OER) and CO₂ reduction reaction (CO₂RR).^{17–21} Nevertheless, the exploration of HEAs in environmental catalysis, particularly for organic pollutant remediation *via* Fenton-like reactions, receives increasing attention. For instance, the FeNiCoCrY HEA nanopowders synthesized *via* liquid-phase reduction exhibit sustained and efficient degradation performance of high-concentration Congo red and Cr(VI) in wastewater, owing to their high specific surface area.²² Dissimilarly, the high-entropy carbide nanoparticles prepared through combustion synthesis achieve approximately 65.5% tetracycline (TC) removal over ten hours *via* persulfate activation.²³ Moreover, the CoCrFeMnNi HEA nanoparticles supported on activated carbon demonstrate excellent degradation behavior of methylene blue without added peroxides, attributing to synergistic effects of solid-solution.²⁴ Despite these advances, the catalytic efficacy of such nanomaterials relies heavily on size-dependent effects, while the absence of scalable and stable synthesis methods continues to hinder their practical application in sustainable wastewater treatment.

Notoriously, the gas atomization and mechanical ball milling will offer a scalable and effective alternative for the engineering-scale manufacturing of HEA powders.^{25,26} In contrast, the rough surfaces of ball-milled powders provide more active sites than the smooth surfaces of gas-atomized powders, which may be beneficial for catalytic reactions.^{27–29} However, the direct ball milling of elemental powder blends frequently leads to issues such as severe compositional

^aSchool of Metallurgy, Northeastern University, Shenyang 110819, China^bInstitute of Corrosion Science and Technology, Guangzhou, 510530, China. E-mail: bwzhao@icost.ac.cn^cXi'an Drainage Management Center, Xi'an, 710018, China

segregation and incomplete alloying, as well as the formation of non-equilibrium phases, rather than a single-phase solid solution with the desired stability.^{30,31} On this basis, whether we are suggested to strategically regulate the ball-milling precursor materials may be able to obtain high activity HEA powder with uniform composition, making high yield and economically feasible preparation.

In summary, the present study proposes a novel integrated strategy combining single-roller melt-spinning with subsequent ball-milling to enable the scalable fabrication of HEA powders. Firstly, we established a convenient and efficient preparation route for FeCoNiCuAl HEA ribbons with homogeneous and tunable compositions *via* melt-spinning and followed by mechanical ball-milling to yield fine powders. Secondly, the HEA powders were comprehensively characterized in terms of particle size distribution, compositional uniformity, and microstructural integrity, aiming to elucidate the phase evolution and morphological transformation mechanisms. Next, the catalytic performance of TC was been evaluated for ball-milled HEA powders as heterogeneous catalysts for activating reactive oxygen species in a Fenton-like system, investigating the dynamic surface evolution and underlying catalytic mechanisms. By accomplishing these objectives, this work not only demonstrates the feasibility of the integrated melt-spinning and ball-milling approach in overcoming the limitations of traditional manufacturing, but also provides a practical and scalable pathway toward the sustainable remediation of TC wastewater using compositionally reliable HEA catalysts.

2. Experimental

2.1. Preparation of HEA powders

The FeCoNiCuAl alloy ingots with a nominal equiatomic composition were manufactured by arc melting under a Ti-gettered argon atmosphere, using raw metallic elements each with an exceeding purity 99.9%. The brittle HEA ribbons were subsequently fabricated from the ingots *via* single roller melt-spinning under argon atmosphere protection.³² The HEA powders were prepared by mechanically ball-milling the shredded alloy pieces for 90 min at two various rotational speeds of 200 rpm and 280 rpm, denoting as BM-200 and BM-280, respectively. The ball-to-material was controlled within a fixed weight ratio of 20 : 1 using steel balls for two different diameters of 8 mm and 6 mm.³³ For comparative purposes, the commercially available spherical FeCoNiCuAl HEA powders produced by gas atomization were purchased from Beijing Yijin New Material Technology Co., Ltd, which were called after GA-

FP and GA-CP according to respective particle sizes, respectively (Fig. 1).

2.2. Characterization techniques

The phases of the powders were identified by X-ray diffraction (XRD, Rigaku D/max 2400, America) using Cu K α radiation. The surface morphology, particle size and component distribution were characterized using a scanning electron microscope (SEM, Zeiss SUPPA 55, Germany) equipped with an energy dispersive spectrometer (EDS). The chemical states and electronic structures were analyzed by X-ray photoelectron spectroscopy (XPS, Thermo ESCALAB 250XI, America) with an Al K α radiation. The microstructural characteristics of the reacted catalysts were observed by transmission electron microscopy (TEM, FEI Talos F200, America) equipped with energy dispersive X-ray spectroscopy (EDS) for elemental mapping. The specific surface area of four powders was measured by Brunauer–Emmett–Teller (BET, Micromeritics APAP 2020, America) technique.

2.3. Degradation method of TC

The catalytic degradation performance of TC (Shanxi Tond Chemical CO., Ltd) was evaluated by the as-prepared FeCoNiCuAl HEA powders. The Fenton-like reactions were conducted in a batch reactor placed in a thermostatic water-bath shaker (SHA-BA) operating at 170 rpm and controlled temperatures of 25 °C, 35 °C and 45 °C. In a typical experiment, 0.5 g of catalyst, 40 mg per L TC and 8.16 mmol per L H₂O₂ were added to 60 mL of TC solution adjusted to pH 3. At 5 min intervals, approximately 2 mL of the reaction mixture was extracted and centrifuged at 10 000 rpm for 2 min, and the supernatant analyzed by UV-vis spectroscopy (Sunny Optical Technology UV-2000, China) and liquid chromatogram-mass spectrometer (LC-MS, Thermo Scientific-Ultimate 3000 HPLC, America). The absorption intensity at $\lambda_{TC} = 358$ nm was recorded to determine the residual TC concentration. The degradation efficiency (η) and reaction rate (k) were calculated according to the following equations for $\eta = (1 - C_t/C_0) \times 100\%$ and $k = \ln(C_0/C_t)/t$, where C_0 and C_t is the initial and residual TC concentrations, respectively. The active radicals were detected by electron paramagnetic resonance spectrometer (EPR, Bruker-EmxPlus, Germany) through DMPO as spin-trapping agent.

3. Results and discussion

3.1. Characterization of HEA powders

The crystalline phases of the as-prepared BM-280, BM-200, GA-FP and GA-CP HEA powders was assessed by XRD patterns.



Fig. 1 Schematic diagram of the synthesis of FeCoNiCuAl HEA powders by ball milling process.

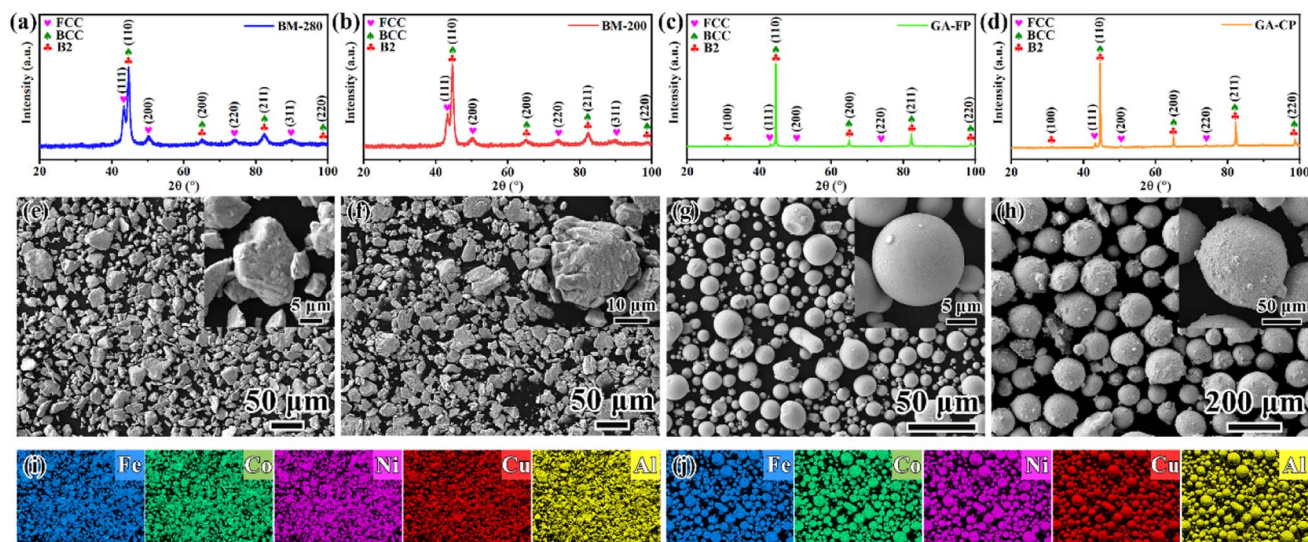


Fig. 2 (a–d) XRD patterns and (e–h) SEM images of four original FeCoNiCuAl HEAs. (i) and (j) Elemental mapping images of (e) and (g).

Emerging from Fig. 2(a–d), all four powders display diffraction peaks at identical positions, corresponding to a dual-phase solid-solution structure comprising both body-centered cubic (BCC) and face-centered cubic (FCC) phases, along with detectable B2-type ordered phase. Specifically, the BCC and B2 phases can be indexed to the (100), (110), (200), (211) and (220) crystal planes, whereas the FCC phase is assigned to the (111), (200), (220) and (311) crystal planes.^{34–36} Notably, the diffraction peaks of the BM-280 and BM-200 powders are considerably broader than those of the GA-FP and GA-CP powders, indicating lower crystallinity and finer coherent scattering domains owing to severe plastic deformation during intensive mechanical milling. Furthermore, the relative intensity of the FCC phase more pronounced in the ball-milled powders, suggesting the distinction of phase distribution and microstructural evolution influenced by two different synthesis pathways.

The morphologies of the fresh BM-280, BM-200, GA-FP and GA-CP HEA powders are observed from Fig. 2(e–h). It can be proved that the particles are uniformly dispersed without agglomeration phenomenon among all types of powders. Explicitly, the GA-FP and GA-CP HEA powders demonstrate spherical features, and the GA-FP powders with 1–25 μm possess a smooth and round surface, whereas the GA-CP powders with 50–120 μm appear some minimum satellite balls. In contrast, the BM-280, BM-200 HEA powders present markedly irregular shapes and highly roughened surfaces, and abundant asperities and nanoscale irregularities are conducive to expanding specific surface area and exposing active sites. Notably, the particle sizes are comparable to those of the GA-FP sample. Furthermore, EDS elemental mapping (Fig. 2(i and j)) confirms that both BM-280 and GA-FP powders consist of Fe, Co, Ni, Cu, and Al elements. All constituent elements uniformly distributed throughout the observed particles, indicating excellent compositional homogeneity.^{37,38} Additionally, the specific surface area of four powders was determined by BET analysis, which are summarized as approximately $2.77 \text{ m}^2 \text{ g}^{-1}$ (BM-280), $1.88 \text{ m}^2 \text{ g}^{-1}$ (BM-200), $2.68 \text{ m}^2 \text{ g}^{-1}$ (GA-FP) and 1.93

$\text{m}^2 \text{ g}^{-1}$ (GA-CP), respectively. It indicates the increase of the specific surface area of powders by high energy ball milling process.

3.2. Catalytic performance of HEA powders

A comprehensive comparison of the degradation capabilities of four HEAs catalysts was evaluated by decomposing TC pollutants across a temperature range from 25 $^{\circ}\text{C}$ to 45 $^{\circ}\text{C}$. As illustrated in Fig. 3(a), it is clearly that the ball-milled powders (BM-280 and BM-200) exhibit significantly superior degradation efficiency compared to their gas-atomized counterparts (GA-FP and GA-CP) under all temperature conditions. Notably, a slight difference appears in the TC removal between BM-200 and BM-280 catalysts. The TC removal efficiency of GA-FP is much higher than that of GA-CP, attributing to the fact that smaller dimensions offer a greater specific surface area. The degradation reaction follows the pseudo-first-order kinetics. Fig. 3(b) further compares the apparent reaction rate constants of the four catalysts as a function of temperature. The degradation rates increase consistently with rising temperature, reflecting the thermally enhanced molecular diffusion and collision frequency that promote reaction kinetics. Among all samples, BM-280 catalysts demonstrate the relatively most outstanding degradation rate, highlighting its optimal catalytic activity.

To further evaluate the degradation capability of the BM-280 catalysts, the influences of H_2O_2 concentration, catalyst dosage and pH value on TC removal efficiency are systematically investigated.^{39,40} As shown in Fig. 3(c), increasing H_2O_2 concentration leads to a marked enhancement in TC degradation. In the absence of H_2O_2 , the degradation efficiency reaches only 49.51% within 50 min, whereas it significantly increased to 92.69% upon the addition of 8.16 mol per L H_2O_2 . This substantial improvement confirms that H_2O_2 serves as the primary source for generating reactive oxygen species ($\cdot\text{OH}$ and $\cdot\text{O}_2^-$), which are essential for driving the Fenton-like reaction.

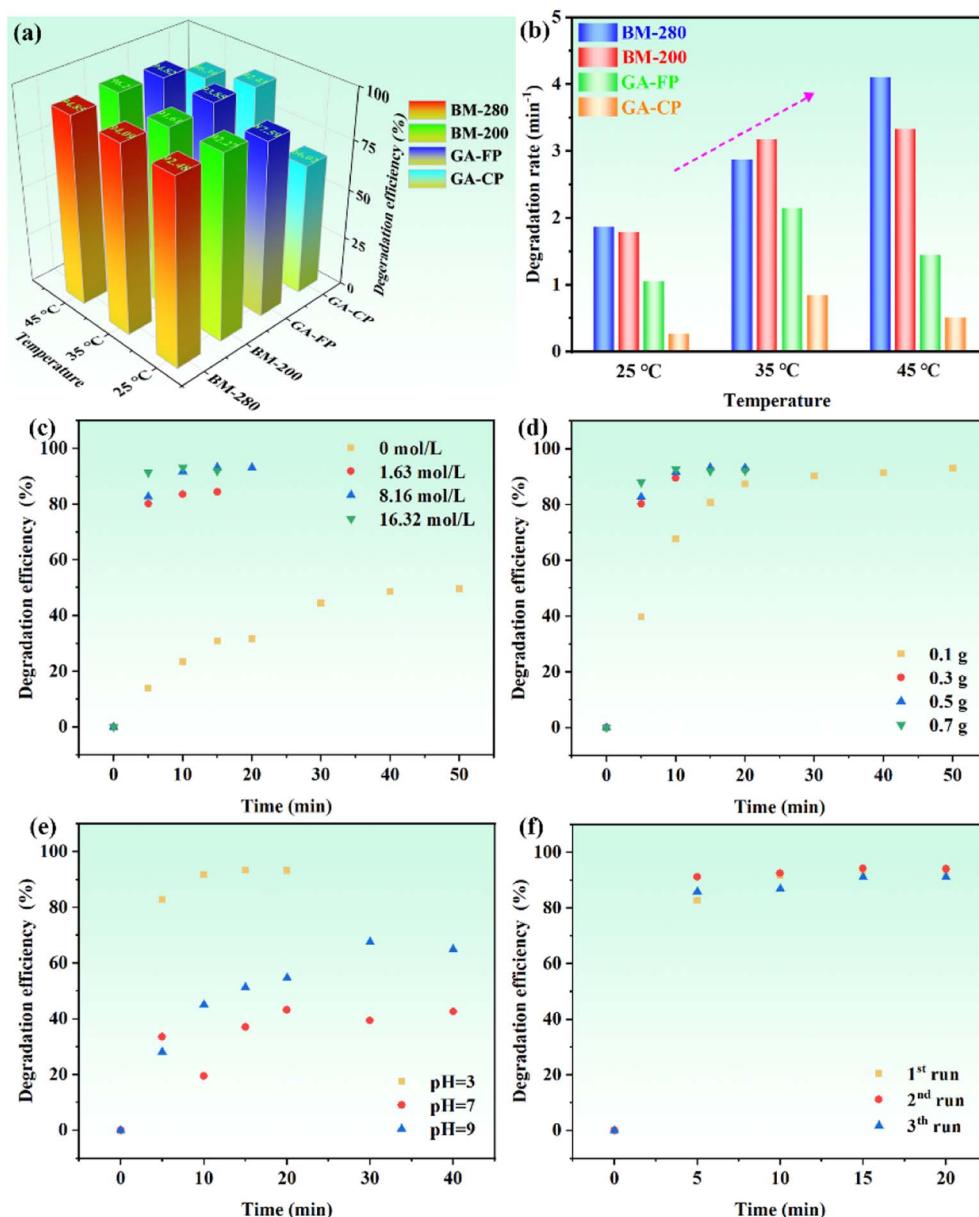


Fig. 3 (a) Degradation efficiency and (b) degradation rate of four HEAs at various temperatures. Effects of (c) H_2O_2 concentration, (d) catalyst dosage, (e) pH values on the degradation efficiency of four HEAs. (f) Reusability test.

As the powder dosage increased from 0.1 to 0.7 g in Fig. 3(d), the degradation efficiency improves remarkably, which can be attributed to the provision of a greater number of accessible active sites that facilitate the catalytic reaction. Notably, the TC solution was almost completely degraded within 20 min at pH = 3 as shown in Fig. 3(e), exhibiting considerably higher efficiency than that observed under neutral or alkaline conditions. This pH-dependent behavior indicates that acidic environments are more favorable for the *in situ* generation of reactive species, thereby promoting the oxidative degradation process. The cycling stability of the BM-280 catalyst was evaluated through consecutive reuse experiments, as depicted in Fig. 3(f). Both the degradation efficiency and rate remained relatively stable over repeated cycles, with the efficiency still exceeding 91.18% even after the third run. It suggests that BM-280 powder presents the

outstanding cycling stability and long service life. Therefore, the BM-280 catalysts provide more active sites and highly irregular surface morphology, thereby creating large numbers of activated free radicals for subsequent efficient mineralization of TC into environmentally friendly small molecules.

3.3 Structures and morphologies of reacted HEA powders

The crystal structure characteristics have no obvious changes for BM-280 and GA-FP HEA powders before and after the reaction as shown in Fig. 4(a), maintaining the BCC, FCC solid-solution phases and B2 phase. The reaction only occurs on the surface of the particles, consuming the active atoms on the surface without altering the stable matrix structure.

The catalytic degradation is normally regarded as a typical interfacial reaction process. The chemical compositions of the surface towards the fresh and reacted BM-280 and GA-FP HEA powders are determined by XPS analysis.^{41,42} In the Fe 2p_{3/2} in Fig. 4(b), compared with the ball-milled powder and gas atomized powder before the reaction, the peaks at 707.03 eV, 709.90 eV and 712.41 eV can be attributed to the presence of Fe⁰, Fe²⁺ and Fe³⁺ for GA-FP HEA catalysts. The peaks shift towards higher binding energies due to ball milling, reaching 707.71 eV, 710.72 eV and 713.30 eV for BM-280 catalysts. After the reaction, the positions of the diffraction peaks corresponding to two powders have no significant changes, but the peak areas of the high valence state characteristic peaks all expended noticeably, indicating the increasement of relative content for oxidized state substances. As depicted in the Fig. 4(c), the Co 2p_{3/2} spectrum can be divided into three characteristic peaks, corresponding to Co⁰, Co²⁺ and Co³⁺ mixed states for as-received and reacted BM-280 and GA-FP HEA catalysts. Following the reaction, the relative content of the oxide corresponding to the high-valent cobalt species increased remarkably owing to redox reaction. Emerging from Ni 2p_{3/2} of Fig. 4(d), the mixed presence of elemental for Ni⁰

and oxidation state for Ni²⁺ appear on the surface of the fresh BM-280 and GA-FP HEA catalysts, the proportion of Ni²⁺ state is enhanced as the ball milling process progresses. For the reacted powders, the peaks shift towards higher binding energy, leading to bonding with atoms carrying a greater electronegativity due to the reduction in the density of the surrounding electron cloud. According to Cu 2p_{3/2} of Fig. 4(e), the binding energy at 932.29 eV and 934.23 eV corresponds to Cu⁰/Cu⁺ states and Cu²⁺ state for GA-FP HEA catalysts, respectively. Similarly, the metallic Cu and oxidized Cu₂O/CuO exist in the surface of the BM-280 catalysts, the relative content proportion of high-valent state substances has increased significantly. In contrast, the content of oxidized copper obviously increased after the reaction, forming a mass of Cu-based nano oxide particles at the interface. In the Al 2p in Fig. 4(f), Al compounds mainly include Al⁰ (74.46 eV) and Al₃O₂ (77.23 eV) on the surface of GA-FP HEA powders and Al⁰ (74.16 eV) and Al₃O₂ (77.18 eV) on the surface of BM-280 HEA powders before the reaction. After the reaction, Al₂O₃ oxide layers slightly increase on the surface. Simultaneously, it is evident that the same substances are present on the surface of the catalysts prepared by gas atomization and ball milling associated to the

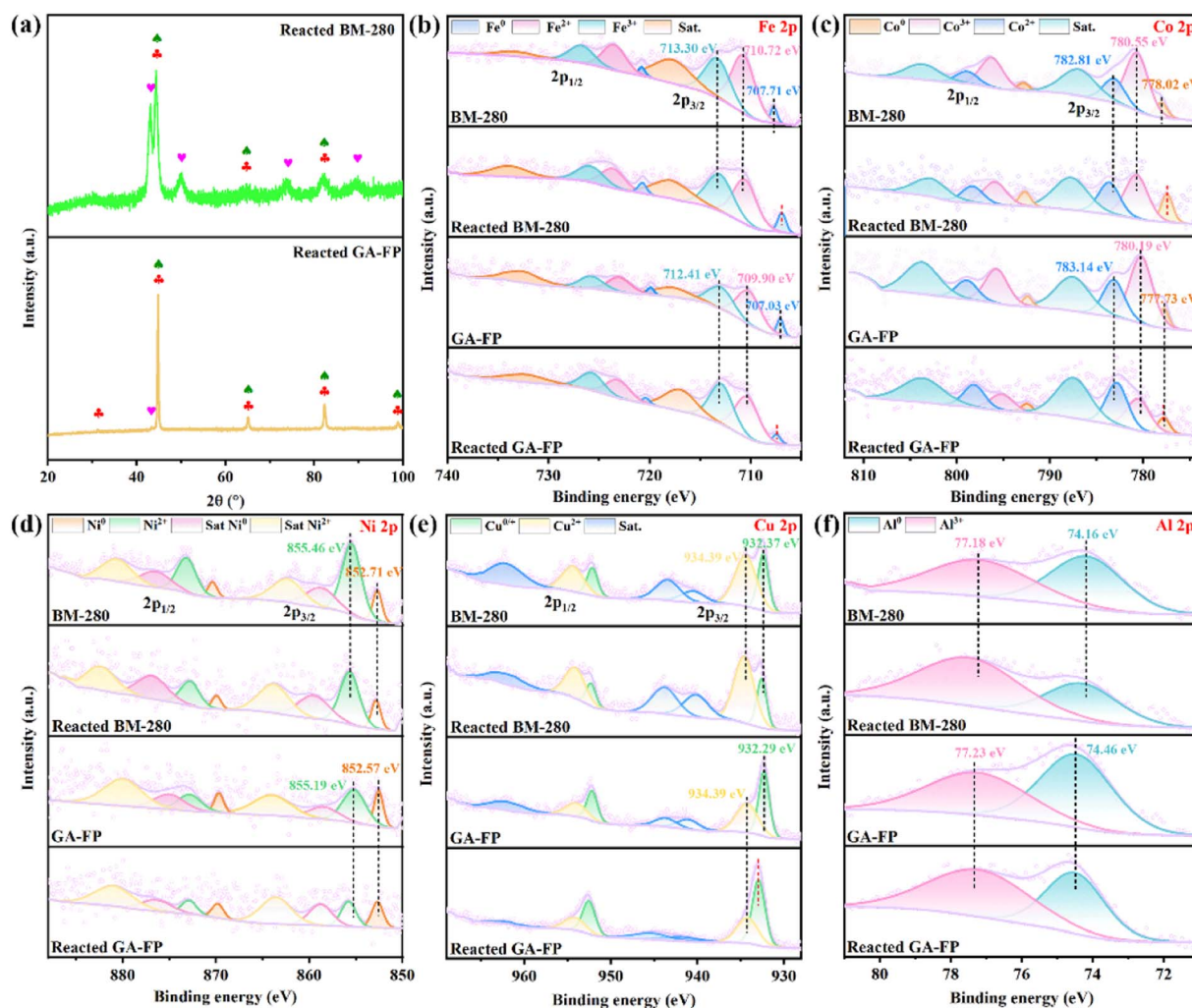


Fig. 4 (a) XRD pattern and XPS spectra of (b) Fe 2p, (c) Co 2p, (d) Ni 2p, (e) Cu 2p and (f) Al 2p in binding energy regions for reacted BM-280 and GA-FP HEA powders.

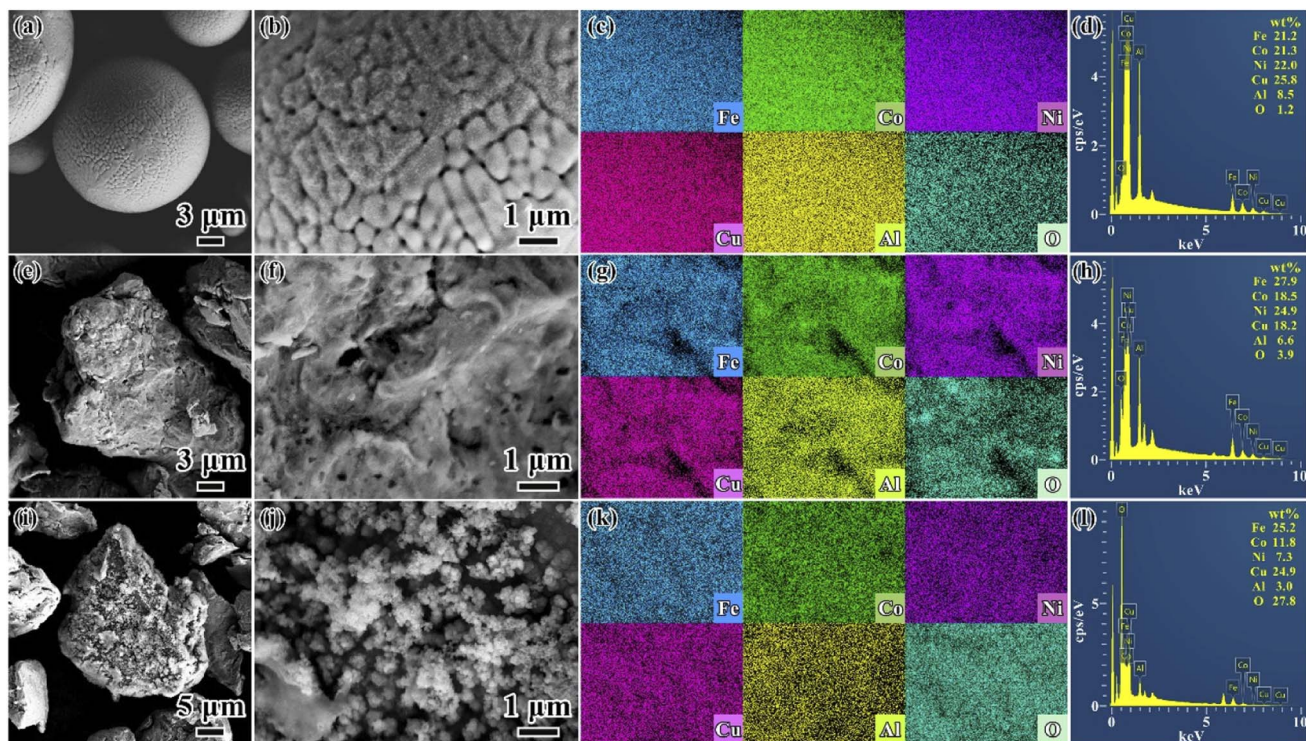


Fig. 5 SEM images and EDS mappings of surface for (a–d) reacted GA-FP HEA powder, (e–h) reacted BM-280 HEA powders and (i–l) another reacted BM-280 HEA powders.

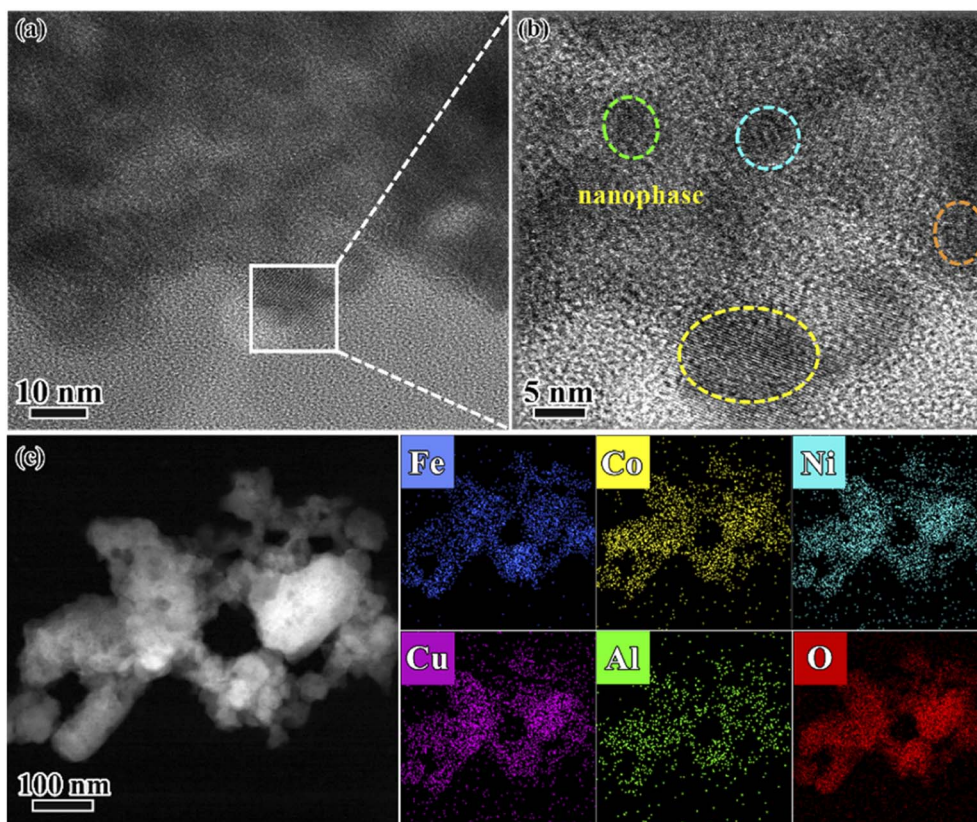


Fig. 6 (a) and (b) HRTEM images and (c) HAADF image and the corresponding elemental mapping images of the surface of the reacted BM-280 HEA powders.

types of characteristic peaks. However, the peak position shifts towards a greater binding energy by ball milling, indicating enhanced surface electron deficiency and oxidation propensity. The formation of oxidized surface species establishes a nano-structured oxide-rich interface, likely contributing to the improved catalytic activity of the BM-280 catalyst.

The surface evolution of between BM-280 and GA-FP HEA powders was observed from the SEM images after the reaction. As seen in Fig. 5(a) and (b), the tenuous corrosion products like a fluffy texture appear on the surface of GA-FP HEA powders, suggesting limited participation of surface-active species during the reaction. In contrast, the ball-milled BM-280 catalyst exhibits apparent surface restructuring. The irregular protrusions and corrugations can be described everywhere on the surface in Fig. 5(e) and (f), while the other situation is that the exaggerated corrosion have appeared on the surface to construct a nanoporous structure based on *in situ* modification. Within these pores, a high density of nanoparticles is densely adhered to the pore walls and interiors in Fig. 5(i) and (j), effectively creating a hierarchical active interface. Corresponding EDS elemental mapping confirms a uniform spatial distribution of the constituent metals (Fe, Co, Ni, Cu, Al) as well as adsorbed oxygen across the analyzed regions for both catalysts.

Quantitative elemental analysis in Fig. 5(d), (h) and (l), while the surface composition of the reacted GA-FP powders remains close to the nominal bulk stoichiometry, the surface of the BM-280 powders demonstrates significant enrichment in Fe, Cu and O elements. This enrichment is particularly pronounced within the nanoporous regions. This localized concentration gradient is consistent with the preferential surface oxidation during the reaction, leading to the formation of nanoporous framework and the precipitation of nanoscale composites. Such a synergistic surface evolution and a porous network can be able to positively expand active site regions and provide electronic transmission channels to promote the catalytic reaction, thus enhancing the degradation performance.^{43,44}

The surface microstructure and compositional evolution of the reacted BM-280 HEA particles are investigated. High-resolution TEM (HRTEM) images Fig. 6(a) and (b), reveal well-defined lattice fringes with distinct interplanar spacings, confirming the presence of nanocrystalline phases formed *in situ* during the catalytic process. The corresponding dark-field scanning TEM image and energy-dispersive X-ray spectroscopy elemental maps in Fig. 6(c) demonstrate a homogeneous spatial distribution of Fe, Co, Ni, Cu, and Al elements across the observed particle surface. The formation of complex nanophase

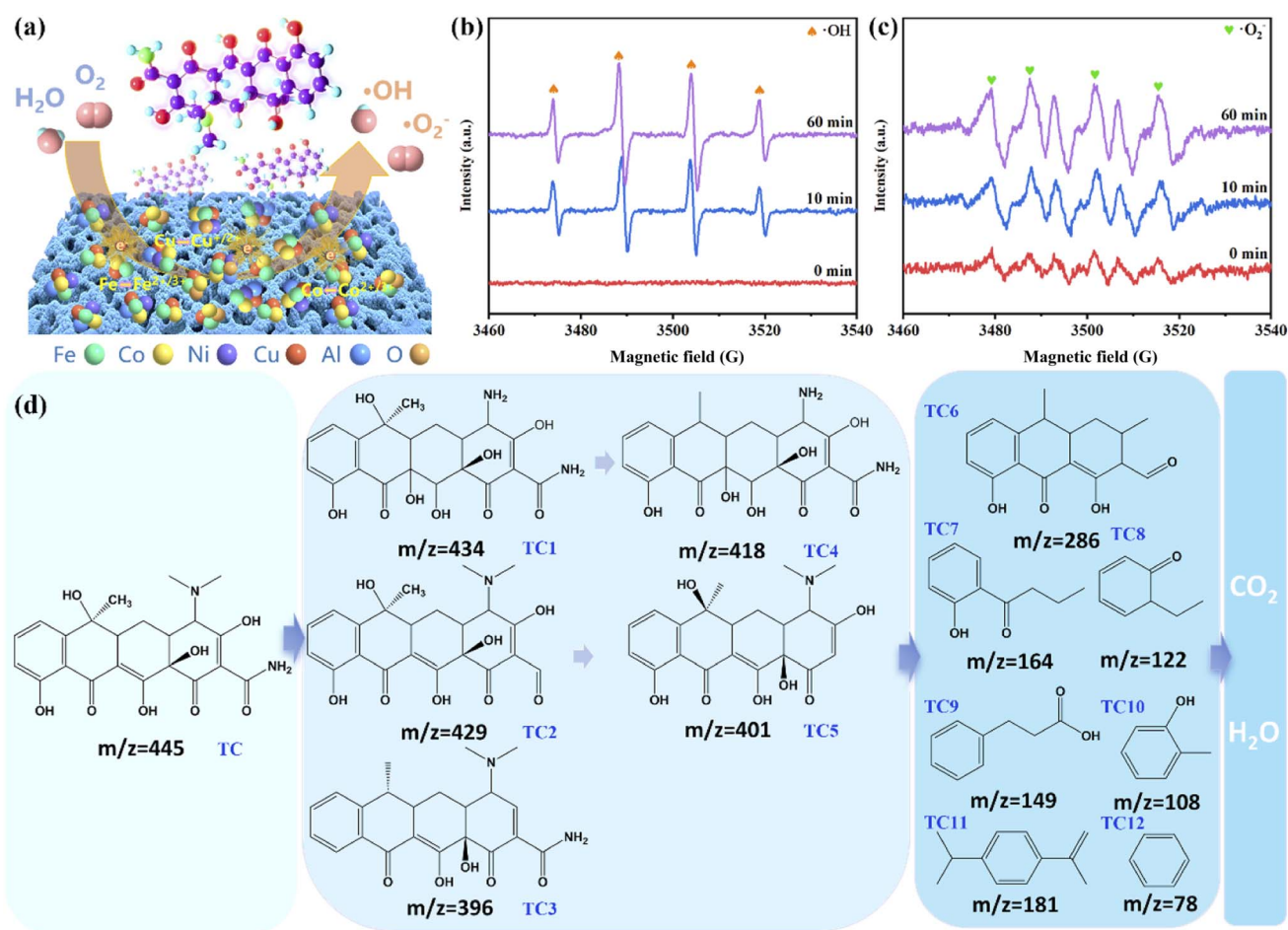


Fig. 7 (a) Schematic of the electrocatalytic reaction process and mechanism, (b) EPR spectra for the detection of (c) DMPO-·OH and (c) DMPO-·O₂⁻ radicals, (d) probable reaction pathways.

mixtures suggests likely comprising metastable metal oxides and high-entropy oxide domains resulting from surface oxidation and reconstruction.^{45,46} Crucially, the development of these multiple nanoscale compounds facilitates the exposure of abundant high-index facets and surface terraces. Such a defect-rich and highly stepped interface is instrumental in optimizing the local electronic structure and increasing the accessible specific surface area. These microstructural features synergistically contribute to improving catalytic performance by promoting reactant adsorption, charge transfer and the provision of a high density of active sites.⁴⁷

3.4 Catalytic mechanism analysis

The possible catalytic mechanism for TC degradation in the HEA/H₂O₂ system can be proposed based on the above comprehensive analysis results, as shown in Fig. 7(a). Initially, the intrinsic multicomponent synergy within the high-entropy solid solution provides a favorable electronic environment, facilitating the adsorption and activation of H₂O₂. During the critical reaction stage, both the metastable surface of the pristine alloy and the *in situ* precipitated nanoscale composites act as highly active sites. These active sites efficiently catalyze the heterolytic cleavage of H₂O₂, generating a flux of potent reactive oxygen species, primarily hydroxyl radicals ($\cdot\text{OH}$) and superoxide radicals ($\cdot\text{O}_2^-$), which rapidly attack and fragment the TC molecules. Emerging from Fig. 7(b) and (c), the typical characteristic peaks of the corresponding $\cdot\text{OH}$ and $\cdot\text{O}_2^-$ radicals were detected in the reaction system through EPR analysis. Concurrently, the nanoporous structure drastically increases the specific surface area and enhances the accessibility of active sites, which can be conducive to the adsorption process of TC molecules. In addition, the residual stress is established within the alloy owing to severe plastic deformation and the atomic energy on the surface of the alloy is improved by mechanical ball milling. The change of residual stress and atomic energy may lower the activation barrier for surface redox reaction, thereby significantly accelerating the overall catalytic kinetics.

Furthermore, the probable degradation pathways and some intermediates of TC by a series of reactions were identified by LC-MS analysis as shown in Fig. 7(d). Firstly, the primitive organic molecular group for TC ($m/z = 445$) is fragmented into TC1 ($m/z = 434$) via demethylation and hydroxylation, TC2 ($m/z = 429$) via deamidation and TC 3 ($m/z = 396$) via demethylation and dehydroxylation due to attacking of active radicals. Besides, TC4 ($m/z = 418$) is obtained by further demethylation and dehydroxylation of TC1. TC2 may transform into TC5 ($m/z = 401$) through demethylation and bond cleavage reaction. Based on deep mineralization, these intermediates further are converted into secondary aromatic compounds by several chemical reactions referring to hydroxylation, oxidation, and ring opening and bond cleavage, viewing as TC6 ($m/z = 286$), TC7 ($m/z = 164$), TC 8 ($m/z = 122$), TC9 ($m/z = 149$), TC10 ($m/z = 108$), TC11 ($m/z = 181$) and TC12 ($m/z = 78$), ultimately mineralized into environmentally friendly CO₂ and H₂O molecules during the degradation process.⁴⁸

4 Conclusion

To evaluating the catalytic performance of two types of HEA powders fabricated by the gas atomization and ball milling, the degradation efficiency and rate were systematically investigated towards degrading TC at variable temperature. Two types of HEA powders indicate excellent degradation efficiency, while the BM-280 powders exhibit preferable degradation capability than others. The enhanced degradation performance of BM-280 powders is attributed to the larger specific surface area as well as more exposed nanoscale active sites based on the surface structure changes caused during the reaction process. This research provides a new and feasible strategy for designing advanced materials toward efficient and sustainable remediation of pharmaceutical contaminants in wastewater.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

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